Daniel Honrales

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Education

Doctor of Philosophy in Computer Science Jan 2025 – Current

The University of Texas at Dallas, Richardson, USA
GPA: 3.852/4.0
Research Focus: *Developing multisensory interfaces that transform human-computer interaction through novel experiences, with a particular emphasis on thermal and tactile haptics. Positioned at the intersection of perceptual science, human-centered design, and immersive systems, this work advances user-centric embodied interaction.*

Master of Science in Computer Science Aug 2024 – Current

The University of Texas at Dallas, Richardson, USA
GPA: 3.852/4.0

Bachelor of Science in Computer Science Aug 2020 – May 2024

The University of Texas at Dallas, Richardson, USA
GPA: 3.910/4.0

Awards and Honors

- **Best Paper Honorable Mention Award** for HeatFlow: A Thermal-Tactile Display for Dynamic 2D Thermal Movements at *ACM UIST 2025*
- **Best Demonstration Award** for Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction at *IEEE WHC 2025*
- **GSA Travel Award** for financial support from Graduate Student Assembly (GSA) to travel and present research work in Fall 2025
- **Best Demonstration Award** for Demonstration of Thermal Flow Illusions with Tactile and Thermal Interactions at *IEEE ISMAR 2024*
- **Best Demo Honorable Mention Award** for Thermal In Motion: Designing Thermal Flow Illusions with Tactile and Thermal Interaction at *ACM UIST 2024*
- **1st Place Winner for PRHI's Big Idea Challenge** for SafeOR at TAMUHack 2023
- **TAMUHack Winner** for OffSight at TAMUHack 2022
- **Collegium V Scholar** for honors program at The University of Texas at Dallas
- **UTD Dean's List 2022, 2023, 2024** for Top 10% GPA among the student body
- **National Merit Scholar** for recognition of promising high school students by the NMSC

Publications

- Haokun Wang, **Daniel Honrales**, Hyunjae Gil, and Jin Ryong Kim, *"Thermal Masking Across the Human Body: Patterns, Pathways, and Perceptual Boundaries*" In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (**CHI 2026**).
- **Daniel Honrales**, Yatharth Singhal, and Jin Ryong Kim, *"Demonstration of HeatFlow: A Thermal-Tactile Display for Dynamic 2D Thermal Movements*" In Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (**CHI 2026**).
- Yatharth Singhal*, **Daniel Honrales***, and Jin Ryong Kim, *"HeatFlow: A Thermal-Tactile Display for Dynamic 2D Thermal Movements*" In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (**UIST 2025**).
- Yatharth Singhal, **Daniel Honrales**, Hsin-Ni Ho, and Jin Ryong Kim, *"Wetness Illusion in Mid-Air*" IEEE

International Symposium on Mixed and Augmented Reality (**ISMAR 2024**).

- Yatharth Singhal, **Daniel Honrales**, and Jin Ryong Kim, *"Demonstration of Thermal Flow Illusions with Tactile and Thermal Interaction*" IEEE International Symposium on Mixed and Augmented Reality (**ISMAR 2024**).
- Yatharth Singhal, **Daniel Honrales**, and Jin Ryong Kim, *"Thermal In Motion: Designing Thermal Flow Illusions with Tactile and Thermal Interaction*" In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (**UIST 2024**).

For a complete list of my publications, please visit my Google Scholar profile.

Work Experience

Research Assistant, MI Lab – Richardson, TX Aug 2025 - Present

- Exploring human-computer interaction theory and applications with haptics, VR, and multisensory experiences
- Investigating thermal-tactile haptic interfaces and human perception mechanisms

Teaching Assistant, The University of Texas at Dallas – Richardson, TX Aug 2025 - Present

- Delivered constructive, user-centric feedback for semester-long application design projects
- Streamlined grading, student correspondence, and class logistics for efficient course organization

Software Development Engineer Intern, Amazon – Seattle, WA May 2023 - Aug 2023

- Built data ingestion pipeline to configure location-based attributes of Alexa out-of-box product experiences
- Developed with Alexa International Team and AWS: Lambda, Step Functions, SQS, SNS, CloudWatch, WorkDocs.

Software Development Engineer Intern, Amazon – Seattle, WA Aug 2022 - Nov 2022

- Built metrics for static and deep neural network model training in Alexa AI NLU
- Developed with Alexa AI team and AWS: Telemetry, Timestream, S3, Lambda, Step Functions.

Projects

HeatFlow 2025

- Developed HeatFlow, a thermal display combining tactile and thermal illusions with a 3x3 actuator array to create dynamic 2D thermal flows.
- Integrated vibration-induced thermal referral and apparent tactile motion to simulate continuous thermal movements, demonstrating its application through immersive VR interfaces and a design tool.
- Demoed at UIST 2025 and CHI 2026

Thermal Motion 2023

- Designed and implemented systems combining thermal referral and apparent tactile motion illusions in a 1D array of vibrotactile actuators and a single thermal actuator to deliver illusory moving thermal sensations.
- Developed software for configuring and designing thermal flows that enhance immersion with VR applications.
- Demoed at UIST 2024 (Best Demo Honorable Mention Award), ISMAR 2024 (Best Demo Award), WHC 2025 (Best Demo Award)

SafeOR 2023

- Operating room monitoring system that improves patient safety by preventing fatigue and surgical mistakes
- Utilizes AI, ML, and NLP (OpenCV, Google's Speech-To-Text, Google Firebase, Flask, and React) to assess visual and auditory input

OffSight 2022

- Web application streamlining file organization and assignment tracking for students in remote learning
- Developed front-end and UI through React, back-end through node.js, express.js, busboy, and Google Firebase

General Experience

Student Volunteer – CHI 2026, UIST 2025, CHI 2025

- Coordinated conference operations and logistics with session support and organizational tasks
- Fostered positive attendee experiences through frontline assistance and people-facing roles

National Merit Ambassador, The University of Texas at Dallas – Richardson, Texas Jan 2023 - May 2024

- Introduced prospective students to UTD campus and life through touring events
- Advocated for school spirit and brand during selection processes

Research Skills

Qualitative and Quantitative Research Methods: Experimental Design, Usability Testing, Heuristic Evaluation, Survey Design, Statistical Methods

Programming: Java, Python, C#, AWS, JavaScript, C, C++, HTML, CSS, React

Development: Git, AWS, VR/AR, Unity, AI Prompt Engineering

Prototyping: Arduino, Raspberry Pi, 3D Printing and Modeling, Photoshop, Figma

Reviewing

ACM Conference on Human Factors in Computing Systems (CHI)	2026
ACM Designing Interactive Systems (DIS)	2026
IEEE EuroHaptics	2026
IEEE Conference on Virtual Reality and 3D User Interfaces (VR)	2026
ACM Symposium on User Interfaces and Technology (UIST)	2026
IEEE EuroHaptics	2024

References

Jin Ryong Kim
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Email: hohsinni@design.kyushu-u.ac.jp